

KC House Price Prediction using Regression

Dataset Used: kc_house_data.csv

This project is based on the King County House Sales dataset and focuses on applying machine learning regression algorithms using the Scikit-learn library. The main objective is to predict actual house prices based on features such as bedrooms, bathrooms, square footage, condition, and grade.

Unlike classification, the price is treated as a continuous variable, making the problem suitable for regression.

Libraries Used

- sklearn.svm → Support Vector Regression (SVR)
- sklearn.tree → Decision Tree Regressor
- sklearn.ensemble → Random Forest & Gradient Boosting
- sklearn.neighbors → KNN Regressor
- sklearn.model_selection → Train-test split
- sklearn.metrics → Model evaluation

Project Objectives

- To predict house prices using regression algorithms
- To implement multiple regression models using Scikit-learn
- To compare model performance using evaluation metrics
- To understand how different regressors perform on the same dataset

Machine Learning Algorithms:

1. Support Vector Regression (SVR)

Definition:

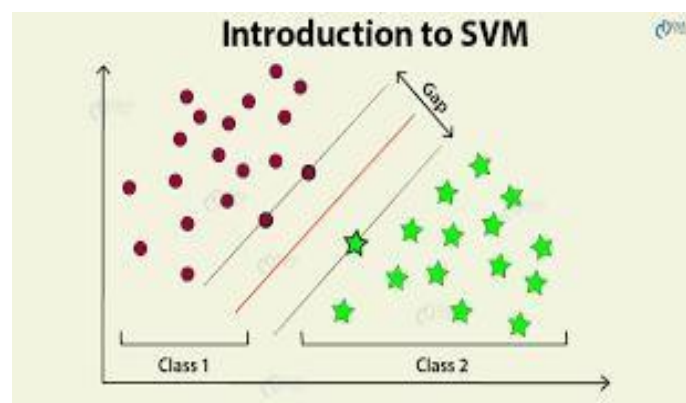
SVR is a regression algorithm that finds a function within a margin of tolerance to predict continuous values.

Use

- Works well with high-dimensional data
- Effective for non-linear relationships

Key Point

- Requires feature scaling



2. Decision Tree Regressor

Definition

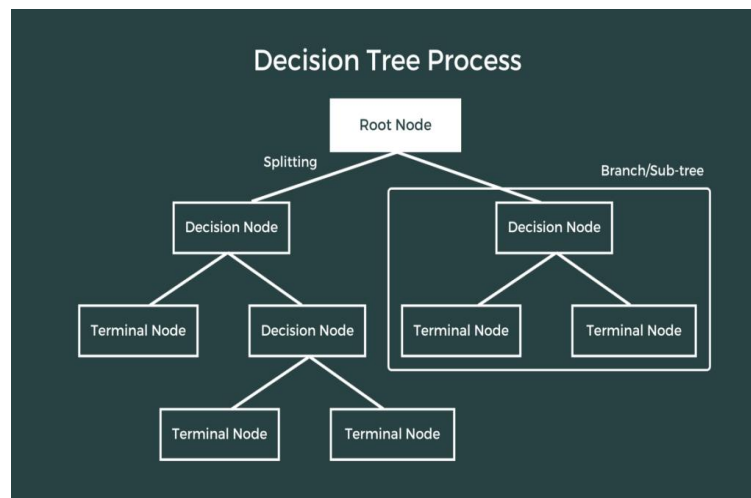
Splits data into branches to predict continuous output values.

Use

- Easy to interpret
- Handles non-linear data

Key Point

- Can overfit



3. Random Forest Regressor

Definition

An ensemble method that combines multiple decision trees to improve prediction accuracy.

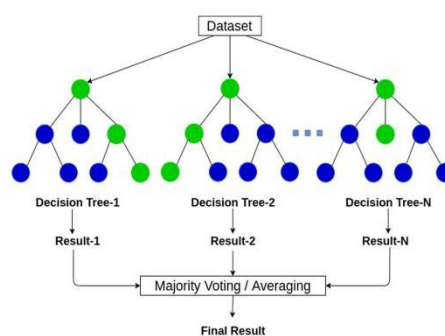
Use

- High accuracy
- Reduces overfitting

Key Point

- Uses multiple trees (bagging)

Random Forest



4. K-Nearest Neighbors (KNN Regressor)

Definition

Predicts values based on the average of nearest neighbors.

Use

- Simple and intuitive
- Works well with smaller dataset

Key Point

- Requires feature scaling

5. Gradient Boosting Regressor

Definition

Builds models sequentially to minimize prediction errors.

Use

- High performance
- Works well on complex datasets

Key Point

- Improves model step by step

6. Linear Regression

Definition

Linear Regression is a simple algorithm that models the relationship between input features and the target variable using a straight line.

Use

- Simple and fast
- Works well with linear relationships

Key Point

- May underperform on complex datasets

7. Extra Trees Regressor

Definition

Extra Trees (Extremely Randomized Trees) is an ensemble method similar to Random Forest but introduces more randomness while building trees.

Use

- High performance
- Faster than Random Forest in some cases

Key Point

- Improves model step by step

8. Ridge Regression

Definition

Ridge Regression is a linear regression technique that adds L2 regularization, which penalizes large coefficients to reduce overfitting.

Use

- Works well when data has multicollinearity
- Helps prevent overfitting in linear models
- Keeps all features but reduces their impact

Key Point

Shrinks coefficients but does not eliminate them

9. Lasso Regression

Definition

Lasso Regression is a linear model that uses L1 regularization, which can shrink some coefficients to zero, effectively performing feature selection.

Use

- Useful when there are many irrelevant features
- Performs automatic feature selection
- Helps simplify the model

Key Point

Can eliminate less important features completely

10.ElasticNet Regression

Definition

ElasticNet is a combination of Ridge and Lasso regression, using both L1 and L2 regularization.

Use

- Works well when features are highly correlated
- Balances feature selection and coefficient shrinking
- More flexible than Ridge or Lasso alone

Key Point

Combines strengths of both Ridge and Lasso

11.AdaBoost Regressor

Definition

AdaBoost (Adaptive Boosting) is an ensemble technique that builds models sequentially, focusing more on correcting errors from previous models.

Use

- Improves performance by focusing on difficult samples
- Works well with weak learners (like small decision trees)
- Reduces bias

Key Point

Each model learns from the mistakes of the previous one

Evaluation Metrics

- **Mean Squared Error (MSE):** Measures average squared error
- **Root Mean Squared Error (RMSE):** Square root of MSE (more interpretable)
- **R² Score:** Measures how well the model explains variance

Conclusion

Different regression algorithms were applied to predict house prices. Ensemble methods like Random Forest and Gradient Boosting generally performed better due to their ability to handle complex relationships. Models like SVR and KNN required feature scaling for better performance. Overall, regression provided more accurate and realistic predictions compared to classification.

What You Have learned

- How to apply regression algorithms to real-world data
- Importance of feature scaling for certain models
- How to evaluate models using MSE, RMSE, and R^2
- Differences between classification and regression problems